

Milestone 4: Final Project Submission + Demonstration

Due: 14 October 2024, Monday, 23:59pm

Marks:90

Instructions

This brief contains the recommended structure and minimum expected content for Milestone 4. It is in every team's interests to adhere to this structure.

The milestone is divided into two (2) components:

Component 1:

MS4 Documentation (due 14 October 2024, 23:59pm)

Component 2:

- The first is a five (5) minute persuasive presentation, i.e. a "sell", of your team's solution.
- The second is a ten (10) minute guided demonstration of the application

MS4 Documentation

Team documentation must include the following:

- Suitable cover page + contents page
- Relevant introduction to the developed application (Description; summarized requirements)
- Updated overall Use Case Diagram, only one use case description (Create Booking), ERD(entity attributes + primary keys and foreign keys), Domain model class diagram, Use Case Realization(Sequence diagram) for create booking (student), update booking (lecturer), update profile either for student or lecturer (admin).
- Report Structure – provide a narrative on the report structure (any one report that will be generated) , included in the application – a design of the report (wireframe or drawn layout using any of the drawing tools or screenshot of the report structure from the application); the tool used for the report in the application (Datagridview or Report Definition Language Client Side [RDLC] Reporting tool in Visual Studio or any other reporting tool included in the application – provide the reasons for selecting the reporting tool)
- Test cases – suitable checklists should be provided to the test scenarios – include test cases for the three use case realization (sequence diagrams) included in the documentation - refer to the test case example provided at the end of this document.
- Peer review form should be either included in your MS4 documentation or submitted with it.
- Team sign-off

Presentation Content

It is customary at the end of a major project to reflect on the work done, identify project strengths and weaknesses and to look to the client's future.

This is your team's final chance to convince the client that the solution being handed over really does meet the requirements and solves the "real world" business problem. It is therefore a persuasive presentation, i.e. an active "sell", rather than a passive report-back.

In order to achieve this, a PowerPoint presentation of approximately five (5) minutes is required. The presentation should include the following (minimum of five slides):

- Overview of the expected benefits of the project to the client, both short- and long-term
- Project highlights, lowlights and lessons learnt (this is your team reflecting on its experience and professional practice)
- Use Case Realizations being demonstrated (Select any two from the documentation)

- ERD

The presentation should be conducted in a professional manner. Factors such as presentation skills, time management, design of slides, overall professionalism etc. will be taken into account.

Note: In a “real world” project environment, only the best speakers in each team would typically make the persuasive presentation. You may choose to follow a similar approach, although a team mark will still be allocated.

Demonstration Content

It is vital that each team embraces this role-play situation and pitches its demonstration appropriately.

Each team will have ten (10) minutes to demonstrate their implementation.

Use cases that **MUST** be demonstrated are as follows:

- Create booking – student view
- Update booking – lecturer view
- Update (student or lecturer profile) – admin view
- Any other feature you would want to showcase

Note: The teams should demonstrate as much of the application as possible

Additional implemented use cases may also be demonstrated, but watch the time allocation!

All reporting implementation has been removed from the scope of the implementation unless it forms part of teams’ “X Factor” functionality. Reports can be showcased in the form of Datagridviews or any other reporting tool during the presentation (see instructions below).

Teams should create PowerPoint presentation containing their ERD and at least two (2) sequence diagrams (use case realizations). Teams must prove that their analysis and design feeds directly into the implementation of their use cases. Any discrepancies should be proactively justified to the client (lecturer).

A reminder that your demonstration must be fully-scripted, meaning that you know exactly what goes in and what comes out. Your database will need to be “rolled back” to a point that is ready for demonstration.

Note: It is highly advisable to tell a story during the demonstration. It is much easier for non-technical clients to follow and empathize with. Another reminder that “nonsense” data is NEVER acceptable. Proper data shows off the true value of your information system and its reports.

Peer Review

All teams must complete and sign the peer review form provided with this brief. It should be submitted with the MS4 documentation.

Presentation Dates and Times

Date	Time	Team
15/10/2024 (Tuesday)	14:00 – 14:15	5
	14:20 – 14:35	22
	14:40 – 14:55	21
	15:00 – 15:15	2
	15:20 – 15:35	28
	15:40 – 15:55	12
	16:00 – 16:15	29
	16:20 – 16:35	23
	16:40 – 16:55	35
	17:00 – 17:15	32
	17:20 – 17:35	26

16/10/2024 (Wednesday)	10:00 – 10:15	6
	10:20 – 10:35	24
	10:40 – 10:55	18
	11:00 – 11:15	7
17/10/2024 (Thursday)	08:20 – 08:35	4
	08:40 – 08:55	15
	09:00 – 09:15	10
18/10/2024 (Friday)	10:00 – 10:15	8
	10:20 – 10:35	3
	10:40 – 10:55	27
	11:00 – 11:15	30
	11:20 – 11:35	9
	11:40 – 11:55	13
21/10/2024 (Monday)	12:20 – 12:35	33
	12:40 – 12:55	19
	13:00 – 13:15	25
	13:20 – 13:35	14
22/10/2024 (Tuesday)	14:00 – 14:15	20
	14:20 – 14:35	1
	14:40 – 14:55	17
	15:00 – 15:15	11
	15:20 – 15:35	16
	15:40 – 15:55	31
	16:00 – 16:15	34
	16:20 – 16:35	
	16:40 – 16:55	
23/10/2024 (Wednesday)	10:00 – 10:10	
	10:15 – 10:25	

These presentations will be conducted online due to venue availability challenges. The Teams link will be communicated to you via the Announcements on Ulwazi.

Test Case Number	1
Test Scenario	To authenticate successful log-in to the application
Test Steps	- User navigates to the login page - User enters an email address and password - Clicks on the sign-in
Prerequisites	Valid user name and password
Test Data	Username and password
Values for test data	Username: abcd@gmail.com Password: Abcde
Expected/Intended Results	Upon successful login, the user is directed to the homepage
Actual Results	As expected
Test Status [Pass/Fail]	Pass

Should the test status be 'Fail', then complete the following

Type of Failure:

Solution:

Milestone 3: Implementation Phase

Due: Monday, 23 September 2024 (23:59pm)**Marks: 90****Milestone 3: Format and submission instructions:**

This milestone has one main input, the project case study/narrative. This brief contains the recommended structure and minimum expected content for Milestone 3. It is in every team's interest to adhere to this structure.

The document must be named as follows:

Team_#_Milestone_3.docx (for Microsoft Word documents)

[PDF formats are also acceptable]

Note: # represents your team number

The document must be written in Arial 11pt font, single spaced and must include page numbers.

A professional title page with milestone number, document title, team member names and student numbers and team logo is mandatory.

Note: These naming and formatting conventions are NOT negotiable, so watch your attention to detail.

Milestone 3:**1. System Sequence Diagrams (SSD)****[10 Marks]**

Use the case study/project narrative. The SSD must be based on the following:

- Create Student or Lecturer profile (choose one user to be included in the diagram)

Make sure to include the relevant objects within the SSD. Feel free to include any alt or loop structures as necessary.

Note: Consistency between the SSD and the relevant revised use case set is essential.

Marking Guideline:

Interacting objects	2 marks
Relevant messages	4 marks
Alt/loops	2 marks
Overall logic	2 marks

2. Sequence Diagrams**[30 Marks]**

Use the case study/project narrative. The sequence diagrams should be drawn separately and must be based on the following (there should be three diagrams):

- To create profile (student)
- To make a venue booking (lecturer)

- To update profile (either for student or lecturer – admin)

Marking Guideline:

Create Profile (Student) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks
Make a venue booking (Lecturer) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks
Update profile (Admin) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks

3. Use Case Realization

[30 Marks]

Use the case study/project narrative. The use case realization must be based on the following:

- Create Profile (Student)
- Make a venue booking (Lecturer)
- Update Lecturer profile (either for student or lecturer – Admin)

Marking Guideline:

Create Profile (Student) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks
Make a venue booking (Lecturer) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks
Update profile (Admin) - Interacting objects - Relevant messages - Alt/loop - Overall logic	2 marks 4 marks 2 marks 2 marks

4. Wireframe

[5 Marks]

Create a wireframe design on **ANY ONE** of the interfaces stated below:

- Main menu from Student view + Student registration page
- Main menu from Lecturer view + Lecturer registration page
- Main menu from Admin view + other options available to the admin

Marking Guideline:

Address the creative goals	2 marks
Visual detail and branding	2 marks
Navigation	2 marks
Page purpose	1 mark
Appropriate content	2 marks
Wireframe elements placed properly	1 mark

5. User Interface Design (images of the C# Form):

[5 Marks]

Create user interface designs for **ANY ONE** user of the following:

Student View:

- Initial registration ('create student profile')
- Main Menu
- Choose any one option on the main menu

Lecturer View

- Main Menu
- Initial registration ('create lecturer profile')
- Choose any one option on the main menu

Feel free to include other interface designs other than those mentioned above.

Note: Basically draft the interfaces the lecturer will view and the student will view. You are free to insert any other entities in your design (next, previous buttons, etc.). User interface designs should be simple and indicate any validation features, such as date format, email format, etc. Hand-drawn drafts or screenshots of the menu screens will be accepted.

Marking Guideline:

Problem space - The user (their needs, goals, limitations and context of use)	2 marks
Information Architecture - Visual hierarchy of difference elements on the screen reflect the user goals - How the items are grouped or placed together to generate different effects.	2 marks
UI Design	

- Elements need to be easily found while also communicating to the user - Pleasing to the eye and no usability issues	2 marks
Design Patterns - Consistency in components included in the design (menus, search/text fields, etc.) - Common patterns for the components	2 marks
UI guides the user to where he/she is trying to go	1 mark
Field validation	1 mark

6. Mock Presentation

[10 Marks]

Presentations to take place 25/09/2024 to 04/10/2024. The mock presentations will be conducted online. All group members to be present for the mock presentations. Each group will have 10 minutes to present the following [Select any two user views to demonstrate during the Mock presentation):

- Create venue booking use case realization
- Basic functionality/working prototype on ONE user view (Student or Lecturer or Admin):
 - o Main menu; Create profile; create venue booking – Student
 - o Main menu; Create profile; create venue booking – Lecturer
 - o Main menu; Update student; update lecturer; generate report

Schedule will be shared via Announcements.

Due: Friday, 23 August 2024 (23:59pm)

Milestone 2: Format and submission instructions:

This milestone has one main input (available on Ulwazi), the project case study/narrative (INFO2001 Resources module). This brief contains the recommended structure and minimum expected content for Milestone 2. It is in every team's interest to adhere to this structure.

The document must be named as follows:

Team_#_Milestone_2 *[Word and PDF formats are acceptable]*

Note: # represents your team number

The document must be written in Arial 11pt font, single spaced and must include page numbers (optional – font type and size should BE consistent throughout the document)

A professional title page with milestone number, document title, team member names and student numbers and team logo is mandatory.

Note: These naming and formatting conventions are NOT negotiable, so watch your attention to detail.

Milestone 2: Requirements

1. Normalization

All relevant entities and assumed attributes (at least three) should be considered for normalization. Identify all appropriate keys in your table structure. All rules for the relevant normal forms MUST be followed and included in your explanations. The following must be shown:

- Full dependency diagram (correctly labelled)
- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)

2. Entity Relationship Diagram (ERD)

The ERD must follow the table structures from normalization. Identify all key attributes, include the attributes per entity (at least 3), relationships and cardinalities must be clearly indicated/included.

The following should be adhered to:

- Relevant entities (as per normalization)
- Labeled relationships between entities
- You can choose one of the two notations [Crows Foot Notation or Chen Notation]
- Cardinalities per entity
- Attributes per entity (at least 3)
- Resolve any many-to-many relations (if they are present)

- A brief narrative should be provided for the ERD.

3. Domain Model Class Diagram

A brief narrative (as an introduction to this diagram) must be provided. The following must be followed:

- Should match the entities/attributes from the ERD
- Labelled associations
- Include multiplicities (match the ERD)
- A basic structure including two sections in the rectangle must be shown (class name and attributes) – you have the option of including data types per attribute and including the third section in the rectangle (procedures/methods/functions)
- Generalization
- Aggregation
- Composition

Marking Rubric:

1	Normalization - Full dependency diagram (PKS, PDs,TDs) - 1NF + explanation - 2NF + explanation - 3NF + explanation	[15] 5 5 5 5
2	ERDs - Entities (5) - Correct notation followed - Labelled relationships - Attributes - Cardinalities	[20] 5 1 4 5 5
3	Domain Model Class Diagrams - Labelled classes - Attributes per class - Labelled associations - Correct notation - Multiplicities - Aggregation - Generalization - Composition	[23] 5 5 4 1 5 1 1 1
	<i>Submission guidelines</i> - Overall a neat consistent document with introductions for each relevant section	2

Semester 2 – Block 3 Milestone Deliverables

Due Date: 26 July 2024 [23:59pm]

Marks: 50

Milestone 1: Format and submission instructions:

This milestone comprises of the project narrative. This brief contains the recommended structure and minimum expected content for Milestone 1. It is in every team's interest to adhere to this structure.

The document must be named as follows:

Team_#_Milestone_1.docx (for Microsoft Word documents) - pdf will also be accepted

Note: # represents your team number

The document must be written in Arial 11pt font, single spaced and must include page numbers.

A professional title page with milestone number, document title, team member names and student numbers and team logo is mandatory.

Note: These naming and formatting conventions are NOT negotiable, so watch your attention to detail.

Milestone 1: Revised Use Case Set + CRUD table

1. *Tabular structure indicating the CRUD features. Refer to images in this document as a guideline for format.*

2. *Revised Use Case Set*

Use the case study/project narrative to create a revised use case set. Each use case should be named using the verb-noun phrase convention and the CRUD technique should be applied for verification.

Show the detailed use case sets for the following:

- **Student - Revised "Create Profile" Use Case**

Use the case study/narrative to create a revised use case description for the Create Profile use case.

- **Lecturer - Revised "Update Venue Booking" Use Case**

Use the updated case study/narrative to create a revised use case description for the Update Venue Booking use case.

- **Admin - Revised "Update Student" Use Case**

Use the case study/project narrative to create a revised use case description for the Update Student use case.

Note: These use case descriptions only need to include Name, Descriptions, Actor(s), Pre-Conditions, PostConditions and Flows of Activity. No other sections are required.

3. Revised Use case diagram

Use the case study/project narrative to create a revised use case diagram. The diagram should display the CRUD features (Create/View, Read, Update and Delete). All relevant <<includes>> and <<extends>> relationships between use cases should be shown.

Milestone Weightings:

Milestone 1 (MS1) = 20%

Milestone 2 (MS2) = 30 %

Milestone 3 (MS3) = 20%

Milestone 4 (MS4) = 30%

Design Phase

Implementation Phase

CRUD Tabular Structure Example:

ENTITY/DOMAIN CLASS	VERIFIED USE CASES	RELATED USE CASES
DONATION	Create donation	View application (includes) View donor (includes) Update application (includes) Update donor (includes) View applicant (includes)
	View donation	
	Update donation	View donation (includes)
APPLICANT	Create applicant	
	View applicant	
	Update applicant	View applicant (includes)
APPLICATION	Create application	View applicant (includes)
	View application	
	Update application	View application (includes)
DONOR	Create donor	
	View donor	
	Update donor	View donor (includes)
ADMIN	Create admin	
	View admin	
	Update admin	View admin (includes)
N/A	Generate report	

Example of a Use Case description as a guideline. Follow the structure that aligns to internal processes of the system. Some processes below may not be relevant to the internal system. Preferred format required.

Use Case Set

Use Case Number	SP_UC 1	
Use Case Name	User registration	
Use Case Description	A user registers for the #Stuport service program by clicking on the 'Register' button and completing the registration form. The actor fills in their personal details. After the student/administrator registered successfully, his/her details are added to the database.	
Actors	Student, administrator	
Pre-conditions	PR1. Actor is on the home page (login/ registration page). PR2. If the user is a student, then he/she must be enrolled as a student at the university. PR3. If the user is an administrator, he/she needs a security-key supplied by the system's manager to register. PR4. The email address entered must be a valid Wits address.	
Main Success Scenarios (basic flow of events)	Actor S1. The actor completes the registration form by entering personal details into the following required fields: - User ID number - First Name - Last Name - Email Address - Contact Number - Password - Password confirmation	System S1.1 The system records the details.
	S2. Actor clicks on the 'Register' button.	S2.1 The system verifies and saves details if it is correct .
Alternate Paths (alternate flow)	A1. Actor does not fill in all information fields and clicks the 'Register' button.	A1.1 The system displays an error message.
Trigger	The actor clicks on the register button on the login page.	
Post-condition	PO1. The user has been registered for the #Stuport program.	

Marking Guideline:

1	CRUD Tabular Structure - At least 5 Entities - Related verified use cases - Related use cases	30/2 = 15 5 3x5 = 15 10
2	Use Case Description - Use Case Name - Description - Actors - Preconditions - Main flow - Post condition	21 1x3 1x3 1x3 1x3 2x3 1X3
3	Use Case diagram - Actors <<include>> <<extends>> - Relevant use cases	11 3 2 2 4
	<i>Submission guidelines</i>	3